



科 目：学术规范与研究生论文写作指导 教师：\*\*\*\*\*

姓 名: \*\*\*\*\* 学 号: \_\_\_\_\_

专 业: 电子信息      类 别: 专业

上课时间: 2026 年 5 月至 2026 年 6 月

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| 平时成绩 | 卷面成绩 | 课程综合成绩 |
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重庆大学研究生院制

# Introduction Section Analysis of *DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning*

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## I. Basic information of the literature

- **Paper Title:** *DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning* [1]
- **Authors:** DeepSeek-AI
- **Journal:** Nature
- **Publication Year:** 2025
- **Volume:** 645
- **Issue (Number):** 8081
- **Pages:** 633–638
- **DOI:** 10.1038/s41586-025-09422-z
- **ISSN:** 1476-4687
- **Publisher:** Springer Science and Business Media LLC

## II. Questions and Answers

### A. What's the research area of the paper your selected? (10 points)

This paper works on making large language models (LLMs) better at reasoning. Specifically, it looks at how LLMs can build reasoning skills on their own via reinforcement learning (RL) with barely any human-labeled data, uses multi-stage training to align reasoning models with human preferences, and distills smaller, efficient reasoning models to release publicly.

### B. What's the importance of the research area? (15 points)

Reasoning is the backbone of both human intelligence and useful advanced AI—it's what lets models handle high-value tasks like math problem-solving, logical deduction, and coding. Progress here cuts down the massive compute power needed to train reasoning-capable LLMs from scratch, eliminates the cost and human bias tied to hand-labeling reasoning data, and lets models come up with totally new reasoning strategies that beat human-defined examples. The smaller distilled models also use way less energy, lowering barriers so more people can access powerful AI tools.

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**C. Extract the background information and rewrite it using your own words. (100 words) (25 points)**

Prior LLM reasoning research has two main tracks. Scaling models big enough during pre-training sometimes makes reasoning skills emerge on their own, but that chews through massive, expensive compute. The other popular approach is chain-of-thought (CoT) prompting: either give the model a few example reasoning steps, or just say “let’s think step by step” to get it to break down its work. Post-training on human-written multistep reasoning traces also boosts performance on math and coding tasks, but these common methods have big constraints that limit further progress.

**D. Identify the gap in this research area and rewrite it using your own words. (100 words) (25 points)**

Existing LLM reasoning methods have two core flaws. Most rely on humans to hand-label reasoning traces for prompting or post-training, which is slow to scale up and bakes human biases into the model. They also force models to copy how humans think, so performance is capped at the level of the human examples they’re trained on—they can’t explore better, non-human reasoning pathways. Early RL-based reasoning models also had poor readability, random language mixing, and weak performance on non-reasoning tasks like writing, making them impractical for real-world use.

**E. What’s the key findings of the paper you selected? (100 words) (25 points)**

This paper has several core takeaways. First is the standout performance of DeepSeek-R1-Zero: as proposed in the paper, the model is trained via reinforcement learning with no human prompting, and when tasked with solving math reasoning problems, it spontaneously outputs the line “Wait, wait. Wait. That’s an aha moment I can flag here,” then starts trying out new reasoning paths to work through the problem. This perfectly demonstrates the emergent growth of the model’s reasoning capabilities.

That said, it has notable flaws: poor readability, occasional mixing of Chinese and English in its outputs, and weak performance on non-reasoning tasks. The follow-up DeepSeek-R1 model addresses these gaps via a multi-stage training framework, while its distilled smaller models outperform their original instruction-tuned counterparts.

### **Appendix: Full Text of the Introduction Section**

Reasoning capability, the cornerstone of human intelligence, enables complex cognitive tasks ranging from mathematical problem-solving to logical deduction and programming. Recent advances in AI have demonstrated that LLMs can exhibit emergent behaviours, including reasoning abilities, when scaled to a sufficient size<sup>4,5</sup>. However, achieving such capabilities in pre-training typically demands substantial computational resources. In parallel, a complementary line of research has demonstrated that LLMs can be effectively augmented through CoT prompting. This technique, which involves either providing carefully designed few-shot examples or using minimalistic prompts such as “Let’s

**Table 1 | An interesting ‘aha moment’ of an intermediate version of DeepSeek-R1-Zero**

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**Question:** If  $a > 1$ , then the sum of the real solutions of  $\sqrt{a - \sqrt{a + x}} = x$  is equal to

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Response: <think>

To solve the equation  $\sqrt{a - \sqrt{a + x}} = x$ , let’s start by squaring both ...

$$(\sqrt{a - \sqrt{a + x}})^2 = x^2 \Rightarrow a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \Rightarrow a^2 - 2ax^2 + (x^2)^2 = a + x \Rightarrow x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

**Wait, wait. Wait. That’s an aha moment I can flag here.**

Let’s reevaluate this step by step to identify whether the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let’s square both sides:

$$a - \sqrt{a + x} = x^2 \Rightarrow \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

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The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of RL.

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**Fig. 1 The “aha moment” of this paper (Table 1). It must be emphasized here that the “aha moment” during the proof process occurred completely independently of any human guidance.**

think step by step”<sup>3,6</sup>, enables models to produce intermediate reasoning steps, thereby substantially enhancing their performance on complex tasks. Similarly, further performance gains have been observed when models learn high-quality, multistep reasoning trajectories during the post-training phase<sup>2,7</sup>. Despite their effectiveness, these approaches exhibit notable limitations. Their dependence on human-annotated reasoning traces slows scalability and introduces cognitive biases. Furthermore, by constraining models to replicate human thought processes, their performance is inherently capped by the human-provided exemplars, which prevents the exploration of superior, non-human-like reasoning pathways.

To tackle these issues, we aim to explore the potential of LLMs for developing reasoning abilities through self-evolution in a RL framework, with minimal reliance on human labelling efforts. Specifically, we build on DeepSeek-V3 Base<sup>8</sup> and use Group Relative Policy Optimization (GRPO)<sup>9</sup> as our RL framework. The reward signal is only based on the correctness of final predictions against ground-truth answers, without imposing constraints on the reasoning process itself. Notably, we bypass the conventional supervised fine-tuning (SFT) phase before RL training. This design choice originates from our hypothesis that human-defined reasoning patterns may limit model exploration, whereas

unrestricted RL training can better incentivize the emergence of new reasoning capabilities in LLMs. Through this process, detailed in the next section, our model (referred to as DeepSeek-R1-Zero) naturally developed diverse and sophisticated reasoning behaviours. To solve reasoning problems, the model exhibits a tendency to generate longer responses, incorporating verification, reflection and the exploration of alternative approaches within each response. Although we do not explicitly teach the model how to reason, it successfully learns improved reasoning strategies through RL.

Although DeepSeek-R1-Zero demonstrates excellent reasoning capabilities, it faces challenges such as poor readability and language mixing, occasionally combining English and Chinese in a single CoT response. Furthermore, the rule-based RL training stage of DeepSeek-R1-Zero is narrowly focused on reasoning tasks, resulting in limited performance in broader areas such as writing and open-domain question answering. To address these challenges, we introduce DeepSeek-R1, a model trained through a multistage learning framework that integrates rejection sampling, RL and supervised fine-tuning, detailed in the ‘DeepSeek-R1’ section. This training pipeline enables DeepSeek-R1 to inherit the reasoning capabilities of its predecessor, DeepSeek-R1-Zero, while aligning model behaviour with human preferences through further non-reasoning data.

To enable broader access to powerful AI at a lower energy cost, we have distilled several smaller models and made them publicly available. These distilled models exhibit strong reasoning capabilities, surpassing the performance of their original instruction-tuned counterparts. We believe that these instruction-tuned versions will also greatly contribute to the research community by providing a valuable resource for understanding the mechanisms underlying long CoT reasoning models and for promoting the development of more powerful reasoning models. We release DeepSeek-R1-Zero, DeepSeek-R1, data samples and distilled models to the public as described in the ‘Code availability’ section.

## Acknowledgments

This paper was written using L<sup>A</sup>T<sub>E</sub>X, and we thank AIAA for providing the template. We also express our gratitude to the DeepSeek-AI team for their research work, which has provided us with valuable learning materials.

## References

- [1] DeepSeek-AI, “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning,” *Nature*, Vol. 645, No. 8081, 2025, pp. 633–638. <https://doi.org/10.1038/s41586-025-09422-z>.